

Practical -08

Code:

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    // malloc
    a = malloc(3 * sizeof(int));

    if (a == NULL)
    {
        printf("malloc failed\n");
        return 1;
    }

    a[0] = 10;
    a[1] = 20;
    a[2] = 30;

    printf("malloc: %d %d %d\n", a[0], a[1], a[2]);

    // calloc
    b = calloc(3, sizeof(int));

    if (b == NULL)
    {
        printf("calloc failed\n");
        free(a);
        return 1;
    }

    printf("calloc: %d %d %d\n", b[0], b[1], b[2]);

    // realloc
    a = realloc(a, 5 * sizeof(int));
```

```
bharath_sai_reddy@LAPTOP-1 × Ubuntu-24.04
GNU nano 7.2
// calloc
b = calloc(3, sizeof(int));

if (b == NULL)
{
    printf("calloc failed\n");
    free(a);
    return 1;
}

printf("calloc: %d %d %d\n", b[0], b[1], b[2]);

// realloc
a = realloc(a, 5 * sizeof(int));

if (a == NULL)
{
    printf("realloc failed\n");
    free(b);
    return 1;
}

a[3] = 40;
a[4] = 50;

printf("realloc: %d %d %d %d %d\n",
      a[0], a[1], a[2], a[3], a[4]);

// free
free(a);
free(b);

printf("Memory freed successfully.\n");

return 0;
}
```

Output:

```

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-08$ ./memory
malloc: 10 20 30
calloc: 0 0 0
realloc: 10 20 30 40 50
Memory freed successfully.
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-08$ valgrind --leak-check=full ./memory
==683== Memcheck, a memory error detector
==683== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==683== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==683== Command: ./memory
==683==
malloc: 10 20 30
calloc: 0 0 0
realloc: 10 20 30 40 50
Memory freed successfully.
==683==
==683== HEAP SUMMARY:
==683==   in use at exit: 0 bytes in 0 blocks
==683==   total heap usage: 4 allocs, 4 frees, 1,068 bytes allocated
==683==
==683== All heap blocks were freed -- no leaks are possible
==683==
==683== For lists of detected and suppressed errors, rerun with: -s
==683== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-08$ |

```

Part -2

Code:

```
bharath_sai_reddy@LAPTOP-1 × Ubuntu-24.04
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int *data = malloc(5 * sizeof(int));

    for (int i = 0; i < 5; i++)
        data[i] = i + 1;

    printf("Before fork:\n");
    printf("Parent data address: %p\n", (void *)data);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("fork failed\n");
        free(data);
        return 1;
    }

    if (pid == 0)
    {
        printf("\nChild process:\n");
        printf("Child data address: %p\n", (void *)data);

        data[0] = 100;

        printf("Child modified data[0] = %d\n", data[0]);

        printf("Child data address after modification: %p\n",
            (void *)data);

        free(data);
    }
    else
    {
        wait(NULL);

        printf("\nParent process:\n");
        printf("Parent data[0] = %d\n", data[0]);

        free(data);
    }

    return 0;
}
```

Output:

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-08$ ./cow
Before fork:
Parent data address: 0x5b78009822a0

Child process:
Child data address: 0x5b78009822a0
Child modified data[0] = 100
Child data address after modification: 0x5b78009822a0

Parent process:
Parent data[0] = 1
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-08$ |
```